

FLYWHEEL ENERGY STORAGE SYSTEM AND THEIR APPLICATIONS

By:

H.E. Jordan

J.D. Herbst

R.J. Hayes

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Center for Electromechanics
The University of Texas at Austin
PRC, Mail Code R7000
Austin, TX 78712
(512) 471-4496

Flywheel Energy Storage Systems and Their Applications

Howard E. Jordan, John D. Herbst, and Richard J. Hayes
The University of Texas at Austin
Center for Electromechanics

Abstract

The use of flywheels to store energy is a technology centuries old. The confluence of several modern technologies has resulted in flywheels becoming a viable solution for the needs of the transportation, electric utility, and aerospace industries.

Advancements in power electronics and high performance composite materials plus the development of reliable magnetic bearings have enhanced the energy storage density and the power delivery capability of flywheel systems, enabling them to serve many new applications. This paper discusses several flywheel energy storage applications and explores in some detail the flywheel systems for the Federal Railroad Administration program, "Advanced Locomotive Propulsion System" (ALPS), and a flywheel system providing energy storage for a hybrid electric bus. The unique motor designs required for these systems are also discussed.

Keywords: flywheel energy storage, turbine-electric locomotive, transit bus flywheel, flywheel battery, hybrid electric vehicles

Introduction

A typical flywheel energy storage system consists of the flywheel, a motor/generator, and power electronics to provide an interface between the energy storage element and the rest of the power system. Scientific advances in several technologies have combined to contribute to the resurgence of flywheels as an energy storage source for a wide range of applications.

Composite materials have been developed that combine graphite fibers and epoxy resins into a high strength, lightweight material ideally suited for flywheel energy storage applications. The composite materials can be wound and cured into cylinders that form the flywheel. Multiple cylindrical layers are then assembled to build up flywheels of the desired moment of inertia. High performance composite flywheels available today can operate at rim speeds up to 1 km/s. Their light weight coupled with high performance motor/generators and power electronics allow composite flywheels to achieve power densities of up to 3,300 W/kg for short (3 s) periods of time [1]. The ALPS locomotive flywheel discussed herein operates at a rim speed of approximately 950 m/s and a power density of 800 W/kg based on the rotor mass only. For comparison purposes, a steel flywheel used in power quality applications [2] operates at a rim

speed of 400 m/s and a power density of approximately 300 W/kg based on rotor mass.

The technological advances in power electronics have enabled the technology that has played an important role in expanding the applications for flywheels. Insulated-gate bipolar transistors (IGBTs) rated for thousands of volts and thousands of amperes are now available. These IGBTs have replaced the gate turn-off thyristors (GTOs) formerly used in megawatt-sized power converters. This has resulted in both size and cost reductions for the power converter unit of a flywheel energy storage system.

Magnetic bearing development has paralleled the flywheel system developments, as the magnetic bearing is an essential element in many energy storage applications. Magnetic bearings suspend the rotor in a magnetic field produced by permanent magnets or electromagnetic coils. Active control coils are used to tailor the field to oppose various forces tending to move the rotor off center. These non-contacting bearings virtually eliminate frictional losses on the rotor, minimizing storage losses in the flywheel and improving system efficiency.

One class of applications that has received widespread attention is the use of flywheels as an energy storage device for motor vehicles: automobiles, buses and railroad locomotives. A common strategy for motive applications is to

size the main prime mover to manage the average drive system load and allow the flywheel to provide the added power for acceleration, hill climbing, etc. Many of these applications also feature energy recovery during braking. Since inverters and other types of power converters can transfer energy in two directions, the braking energy can be used to reaccelerate the flywheel. Usually braking energy must be supplemented by energy from the prime mover to keep the flywheel close to its full speed between peak power demand periods. Two such vehicular applications, a locomotive and a bus, are discussed herein.

Many manufacturing operations cannot tolerate electric power interruptions without disastrous results. The flywheel systems can supply the electric power during interruptions in the utility power system, which typically last from a few milliseconds to a few seconds. If longer interruptions must be managed, the flywheel system can be interconnected with a generator powered by fossil fuel. The engine can be started and readied for operation while the flywheel is carrying the load. Continuity of the electric utility supply is the most mature commercial application of flywheel energy storage according to Hebner, Beno, and Walls [2]. They cite an example of a multiple flywheel system rated up to 5 MW operating to smooth power for a semiconductor fabrication facility.

One of the more prodigious examples is a flywheel being developed for use on the International Space Station [2,3]. Development is centered at the NASA Glenn Research Center in Cleveland, Ohio, and has demonstrated speeds up to 60,000 rpm. While operating in sunlight, a brushless dc motor utilizing power supplied by photovoltaic arrays accelerates the flywheel. During periods when the space station is in the shadow of the earth, the flywheel provides electric energy via the motor, now acting as a generator, to supply equipment and life support system needs. The flywheel batteries are expected to have a service life approximately equal to the space station and do not contain any hazardous chemicals, as do the electrochemical batteries that the flywheel system is replacing. Extending the equipment replacement interval for a space station is a significant advantage, reducing both launch and maintenance costs.

Advanced Locomotive Propulsion System (ALPS)

The ALPS program is part of the Next Generation High Speed Rail Program sponsored by the Federal Railroad Administration (FRA). Converting an existing right-of-way to overhead electric power is estimated to cost between \$3-\$5 million per track mile. The goal of the ALPS program is to develop a fossil-fueled locomotive propulsion system capable of sustaining speeds of 150 mph on the existing infrastructure. The weight of conventional diesel electric locomotives can create track damaging dynamic forces at speeds above 125 mph. The prime mover for the ALPS system is a lightweight high speed gas turbine that reduces locomotive weight and dynamic track forces at higher speeds. The ALPS program provides a practical alternative for improving passenger rail transportation at an affordable cost.

The ALPS system has been described by Herbst, et al. [4]. A schematic diagram for the ALPS power system is displayed in Figure 1.

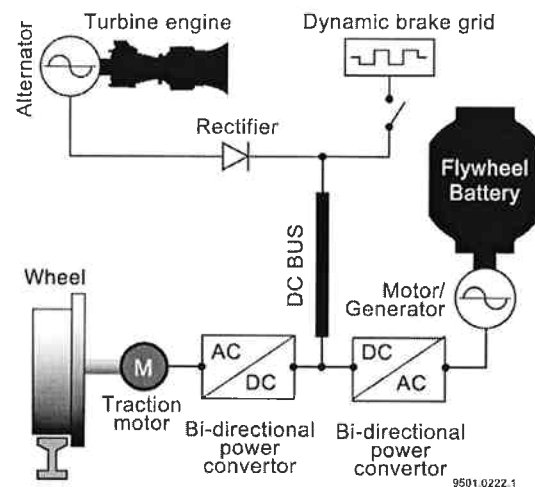


Figure 1. ALPS System schematic diagram

The main propulsion power is provided by a 4,000 horsepower gas turbine driving a synchronous alternator rated at 2 MW with a peak output of 3 MW. The alternator output is rectified to provide power to a dc bus, which distributes power throughout the locomotive. The ac traction motors of the locomotive are powered through variable frequency inverters connected to the dc bus. The flywheel motor/generator is also connected to the dc bus through

a bi-directional, variable frequency power converter.

The design of the ALPS flywheel system is leveraged on experience gained at The University of Texas Center for Electromechanics (UT-CEM) during the development of high energy, high power pulsed power supplies. The ALPS flywheel system is comprised of a 480 MJ energy storage flywheel coupled to a 2 MW, high-speed motor generator. The operating speed range of the flywheel is 7,500-15,000 rpm.

The system is designed to provide up to 3,000 hp for a period of 3 min, with a primary goal of providing power boost for acceleration and speed maintenance on grades. The flywheel also recovers braking energy and can be used to provide load leveling for the gas turbine prime mover. Operation of the turbine at a relatively constant load reduces thermal cycling and significantly extends turbine maintenance intervals. Figure 2 is a view of the flywheel with the major components identified [4].

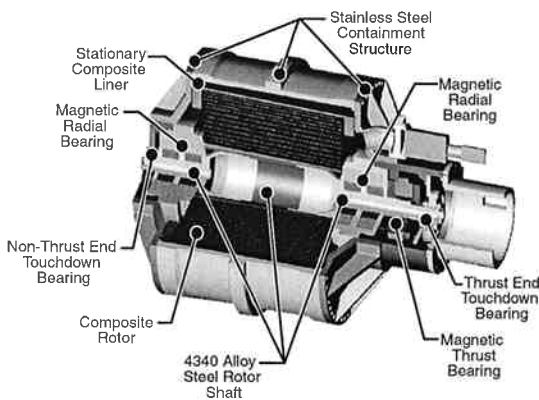


Figure 2. Section view of energy storage flywheel

The flywheel composite rotor is made up of 14 concentric composite rings. The composite rings are filament-wound structures using glass and graphite fibers pre-impregnated with an elevated temperature cure epoxy resin. The concentric rings are assembled with interference fits to manage radial stresses developed during spin.

The rotor is levitated on magnetic bearings to minimize rotational losses and allow long term operation in a vacuum environment. Both the radial and axial magnetic bearings are permanent magnet-bias type with active control coils mounted in the stator. This configuration provides stable control of the rotor in a

locomotive dynamic environment. Ceramic rolling element touchdown bearings are also provided for times when the rotor is not levitated magnetically.

The flywheel operates in a 10^{-3} torr vacuum to minimize windage losses. The flywheel rotor is coupled to an induction motor/generator operating at atmospheric pressure. A Ferrofluid[®] seal maintains the flywheel vacuum.

The induction motor/generator is a copper-bar induction machine rated at 2 MW over the speed range of 7,500 to 15,000 rpm. It is connected through an inverter to the dc bus of the locomotive and can operate in either the motoring or generating regions, depending upon whether it is commanded to accelerate the flywheel or to deliver power to the dc bus. The control operates in a constant V/Hz mode (constant torque) in the 7,500 to 12,000 rpm speed range and in a constant voltage (constant power) mode from 12,000 to 15,000 rpm.

Two unique features of the induction machine design are the rotor construction and the stator winding design.

It is not surprising that centrifugal forces are a major factor in the design of the rotor, considering that the rotor surface velocity is nearly 300 m/s at 15,000 rpm. A round copper bar is used in place of the more usual rectangular bar to provide a better stress pattern in the rotor teeth. Also, a proprietary design for locking the end ring to the rotor structure has been developed to manage the centrifugal forces on the end ring.

At 15,000 rpm, the two-pole armature winding must operate at 250 Hz. This frequency is high enough to cause significant "skin effect" losses in the stator conductors. The final design utilizes 20 rectangular copper conductors (0.045 in. thick) in parallel for each turn to control the ac resistance loss. This design achieved a 1.1 ratio of ac to dc resistance. Earlier designs had ratios of 1.5 and higher.

The benefits of using a flywheel for the high-speed locomotive application were quantified by simulation studies made for the New York to Boston route. Although this particular route is an electrified rail and not a candidate for the locomotive discussed here, it represents a typical intercity route and much data is available from this route for the study. Figure 3 displays the traction system duty cycle developed from the simulation.

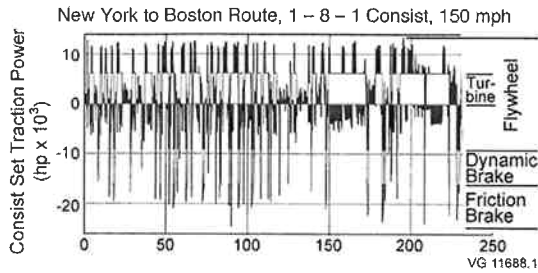


Figure 3. New York to Boston Route duty cycle

As evidenced in figure 3, the flywheel is accomplishing its mission of providing power for accelerations and recovering braking energy. The turbine is, therefore, able to operate at reasonably level loading. A performance comparison [4] concluded that adding a flywheel to the locomotive system resulted in an 8% reduction in fuel consumption, at least a 5 min savings in trip time, and a reduction of maintenance cost on the gas turbine.

At the present time, the flywheel is undergoing spin testing. Both the induction machine and its associated inverter are through the design stages and into manufacturing.

Houston Metro Transit Bus

UT-CEM has also developed a flywheel energy storage system for installation into the Advanced Technology Transit Bus (ATTB) as part of a program sponsored by DARPA and the Houston Metropolitan Transit Authority. While there are many similarities between the transit bus flywheel system and the locomotive system described above, there are also some important differences.

The transit bus system shares a common philosophy with the locomotive in that the flywheel serves as a load-leveling device. The braking energy is used to accelerate the flywheel and the flywheel-stored energy is used to supplement the main power plant to achieve acceleration or to maintain speed on grades. As one might suspect, the energy and power ratings are considerably smaller than the equivalent values for the locomotive. A comparison of key parameters for the two flywheel designs is presented in Table 1.

The transit bus unit is a partially integrated design with both the flywheel and the motor/generator contained in a vacuum chamber. The motor/generator is a permanent magnet, synchronous machine with the permanent magnets mounted on the rotor. A cross sectional view of the transit bus flywheel battery is displayed in Figure 4.

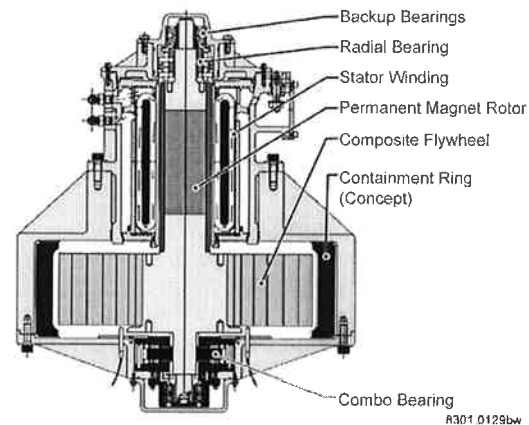


Figure 4. Cross-sectional view of the bus transit flywheel battery [5]

Table 1. Flywheel applications comparison

Parameter	ALPS System	Transit Bus System	NASA Space Station
Energy Stored kW-hr	133	2	3.6
Peak Power kW	2,200	150	3.6
Typical Discharge Time	3 minutes	30 seconds	30 minutes
Rotational Speed rpm	15,000	40,000	53,000
Machine Weight lbs	19,000	450	250
Motor/Generator	Induction	Permanent Magnet	Permanent magnet
Cooling	Air and Oil	Oil and Water	Cold plate
Bearings	Homopolar Magnetic	Homopolar Magnetic	Homopolar Magnetic
Flywheel Design	UT-CEM Cylindrical	UT-CEM Cylindrical	UT-CEM Cylindrical
Rotor Tip Speed m/s	950	935	920

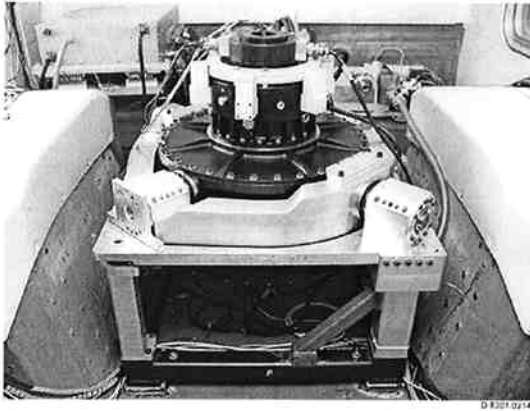


Figure 5. Flywheel battery unit installed in a transit bus.

The stator windings, which are at atmospheric pressure, are liquid-cooled. The rotor is in a vacuum, however, and the heat generated by eddy currents in any part of the rotor must be removed by radiation. The always-present air-gap space harmonic fields, plus the time harmonics caused by inverting switching, act to induce eddy currents in a beryllium-copper sleeve that surrounds the rotor. Significant development effort was devoted to reducing the rotor losses to <100 W [6] so that the losses can be dissipated without excessive temperature rise in the rotor.

The transit bus flywheel system has successfully completed laboratory tests and is presently being field-tested at UT-CEM. A picture of the unit installed in the bus is displayed in Figure 5.

Summary

Flywheels have become a viable solution for the needs of the transportation, electric utility and aerospace industries. Advancements made in power electronics and high performance composite materials plus the development of reliable magnetic bearings have enabled flywheel energy storage systems to advantageously serve certain needs in these and other industries. Specific examples have been presented.

Flywheel energy storage systems are still in the early stages of the development cycle. As improvements in the technology continue, more applications for flywheels as a replacement for batteries, capacitors and other energy storage systems will occur.

Acknowledgement

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